OPERA NUMERORUM

Archive Map -- Reading Order Guide

47 Documents in Canonical Order

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This map shows every document in the Opera Numerorum archive in the order it should be read. Each entry shows the document number, filename, what it contains, and its role in the causal chain. Documents 1-11 are narrative and canonical. Documents 12-43 are machine-certified SHA-bound modules. Documents 44-47 are the Morning Star Engineering series.

Section 1 -- Preface (1 document)

#	Filename	Contents	Role
1	OperaNumerorum_Preface.pdf	Illustrated front-matter. 5 figures: alpha_0 number line, [illegible]	[illegible]

Section 2 -- Original Source Papers (7 documents)

These are David Fox's original written papers for modules M1-M7. They contain the mathematical exposition, proofs, and claims. The corresponding certificate PDFs in Section 5 provide machine verification of every numerical result stated here.

#	Filename	Module	Claim
2	Paper_1_Alpha0_Definition.pdf	M1	alpha_0 = 299 + pi/10 to 5000 decimal places
3	Paper_2_Kappa_Bound.pdf	M2	Kappa bound on the exceptional set (80-bit long double)
4	Paper_3_Q5P5_Bound.pdf	M3	CF of pi/10: Q_5=226, convergent bound=82829
5	Paper_4_S14_Head.pdf	M4	S_14: 14 primes in the exceptional set, p_5 > 82829
6	Paper_5_Bost_Bound.pdf	M5	C(S_4) = 11.4221 > 2*sqrt(13) = 7.2111 (PASS)
7	Paper_6_GRH_X0143.pdf	M6	genus(X_0(143))=13, h(-143)=10, Bost bound satisfied
8	Paper_7_Manifest.pdf	M7	Master manifest over M1-M6; SHA-256 binding

Section 3 -- Canonical Paper (1 document)

#	Filename	Contents	Role
9	Canonical_Paper.pdf	The complete unified paper: exceptional primes for pi/10, [illegible]	[illegible]

Section 4 -- Complete Compiled Documents (2 documents)

#	Filename	Contents	Role
10	OperaNumerorum_Complete.pdf	Full Opera Numerorum compilation: all modules, all proofs, all figures, [illegible] under the Opera Num	The public-facing complete document.
11	BattlePlan_v1.6_Complete.pdf	Same content, internal working title Battle Plan v1.6 preserved to maintain SHA chain integrity.	The internal name. Both names are [illegible] -- one is how it

Section 5 -- SHA-Bound Machine Certification Modules (32 documents)

Each of these PDFs is a machine-generated certificate. The certified stdout of the underlying Python or C program was run in this environment, its SHA-256 hash was computed, and the PDF binds the source code, the output, and the hash together. Any upstream change breaks the chain. Read these alongside the corresponding source papers in Section 2.

#	Filename	Module	Certified Claim
12	Module_1_Certificate.pdf	M1	alpha_0 = 299.314159265... (5000 dps, mpmath 64 dps verified)
13	Module_2_Certificate.pdf	M2	Kappa bound: 80-bit long double, C binary certified
14	Module_3_Certificate.pdf	M3	CF seed corrected; Q_5=226, p_5 bound=82829
15	Module_4_Certificate.pdf	M4	S_14: 14 primes, p_5=83497 > 82829 (PASS)
16	Module_5_Certificate.pdf	M5	C(S_4)=11.4221 > 7.2111=2*sqrt(13). Formula log(p)*p/(p-1) confirmed.
17	Module_6_Certificate.pdf	M6	genus(X_0(143))=13; h(-143)=10 (10 reduced forms enumerated)
18	Module_6_3_Certificate.pdf	M6.3	M6 correction patch: h(-143) audit and Bost bound re-verification
19	Module_7_Certificate.pdf	M7	Master manifest SHA-256(cat m1..m6.out) = 5b80b84d... LOCKED
20	Module_8_Certificate.pdf	M8	rank(H_13(L_w, J_0(143))) = g = 13 (Hankel rank check)
21	Module_9_Certificate.pdf	M9	BSD J_0(143): analytic rank computation
22	Module_9_All_140.pdf	M9-All	Extended S14 check: all 140 primes in set verified
23	Module_10_Genus33.pdf	M10	Genus 33 verification (companion curve)
24	Module_14_S4_Quaternions.pdf	M14	S_4 quaternion structure certification
25	Module_15_Delta_Boost.pdf	M15	Delta boost factor certification
26	Module_16_c_Bridge.pdf	M16	c-bridge between analytic and arithmetic BSD ranks
27	Module_17_Cert_Patch.pdf	M17	Correction patch: five LaTeX draft errors caught and superseded
28	Module_18_Resonance_Ladder.pdf	M18	Resonance ladder structure for X_0(143) Hecke eigenvalues
29	Module_19_p6_Prediction.pdf	M19	p_6 prediction from exceptional prime structure
30	Module_20_p7_Prediction.pdf	M20	p_7 prediction certification
31	Module_21_H4_Invariant.pdf	M21	H_4 invariant certification
32	Module_22_MStar_Definition.pdf	M22	M* = 4/55 definition and rational arithmetic verification
33	Module_23_BSD_J0_143.pdf	M23	BSD conjecture for J_0(143): rank, regulator, Sha bound
34	Module_M8C_ZoeMstar.pdf	M8C	Zoe-M* bridge: Z=15, M*=4/55, 200 Hodge classes transcendental
35	Module_M8D_Resonator.pdf	M8D	120-cell resonator: f_res=alpha_0 MHz, C jumps 5.724x at k_c=3.183
36	Module_M8F_LeanProtocol.pdf	M8F	7-layer protocol: k_eff=3.183, v_g=3.183c, all 8 checks PASS
37	Module_M8G_Provenance.pdf	M8G	Provenance Feb2025->M8F; wormhole=0.524ns; PHS topology
38	Module_M8G_Correction.pdf	M8G-Corr	Z=rank(M_ij) clarification; conditional wormhole cert
39	Module_M8H_G_Amplifier.pdf	M8H	G_eff(Z)=G_0*(Z_vac/Z)^4; A=15^4=50625; F=3.38e-10 N
40	Module_M8I_Wormhole.pdf	M8I	Morris-Thorne wormhole r0=3m; b'=0 PASS; E_cav=1.44 MWh; 14 modes
41	Module_M8J_OQ2_Closure.pdf	M8J	OQ-1: tidal=0.0999g<0.1g; OQ-2: Delta_tau=7.647ns closed
42	Tendon_A_Certificate.pdf	Tendon-A	Structural connector A: bridges M7 manifest to M8 Hankel chain
43	Tendon_B_Certificate.pdf	Tendon-B	Structural connector B: bridges M8-series to Morning Star chain

Section 6 -- Morning Star Engineering Series (4 documents)

Opera Numerorum II. Speculative engineering grounded in certified mathematics. Read after completing Sections 1-5. FTL Certification: MS-FTL-20260523-001.

#	Filename	Contents	Role
44	00_MorningStar_Summary.pdf	10-section overview: causal chain, Phase-Z metric, [illegible]	[illegible]
45	M8K_FTL_Morningstar_TechStack.pdf	3-layer FTL stack: B_M=21.768 MHz, v_g=3.183c, [illegible]	Certified: [illegible] MORNINGSTAR [illegible] CERTIFIED. SHA: 0ae8[illegible]
46	M8L_MorningStar_D20_Operations.pdf	D20 hub: Euler topology, 30 routes, 12 destinations, [illegible]	Certified: MORNINGSTAR_OPERATIONAL_CERTIFIED. [illegible]
47	M8M_MorningStar_Physics_Beyond.pdf	95 routes, daily ops 84tx/512pax/1084.7ly, MTBF=5.50yr, [illegible]	Certified: MORNINGSTAR_PHYSICS_CERTIFIED. [illegible]

Recommended Reading Paths

New reader, start here:

1 (Preface) --> 9 (Canonical Paper) --> 10 (Opera Numerorum Complete) --> 44 (Morning Star Summary)

For the mathematics in detail:

1 --> 2-8 (Source Papers) --> 12-19 (Core certificates M1-M7) --> 20-43 (Extended certificates M8-M23, Tendons) --> 9 (Canonical) --> 10-11 (Complete)

For the Morning Star engineering:

1 --> 9 --> 34-41 (M8C through M8J) --> 44-47 (Morning Star folder)

For independent verification (reproduced from scratch):

Run bash verify_all.sh to reproduce the M1-M6 chain and check the M7 master manifest SHA. Every certificate was produced by running the corresponding Python or C source in this environment.

Master SHA Reference (locked outputs)

Module	Stdout File	SHA-256
M1	m1.out	63ef870a...
M2	m2.out	3716c7db...
M3	m3.out	e687bb09...
M4	m4.out	b810a7a3...
M5	m5.out	9df98a39...
M6	m6.out	ec9fa8c3...
M7 master	cat m1..m6	5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9
M8	m8.out	e2d70821...
M8C	m8c.out	02fe6048...
M8D	m8d.out	27d8e0c1...
M8F	m8f.out	0bd6cee4...
M8G	m8g.out	2874d4bd...
M8G-Corr	m8gc.out	62492d66...
M8H	m8h.out	2c3ac1d2...
M8I	m8i.out	5c7189fc...
M8J	m8j.out	298d440a...
M8K	m8k.out	0ae865a8812ce93b05461ec4483ad1714e24fc9be9de1e7bb54963da43592087
M8L	m8l.out	80ff8a251c6ea7b6a57fd81fe71a76dd62a3f862c80381d571e2f30d3c4222ad

M8M	m8m.out	afce5f2146c40c22bbcc7d7f1c4514eeba08107436de7929a3e3ef6d4f5e121f

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Opera Numerorum -- the Works of Numbers. Every SHA is computed, never invented. Every error is documented, never hidden.